YEAR : 2026 - 2027

RESEARCH PROPOSAL

Presented by: Group 3 - Blk 9

COMPETENCY BASED-LEARNING CHALLENGES ON FORENSIC BALLISTICS

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BACKGROUND OF THE STUDY

Criminology education demands practical competencies, especially in technical subjects like Forensic Ballistics. While Competency-Based Learning (CBL) connects instruction with professional expectations, students still face challenges in course content, laboratory resources, activities, and instructional support. Despite existing local studies, there is a lack of institution-specific data at PHILCST. Thus, this descriptive-quantitative study aims to identify the learning challenges of fourth-year Criminology students in Forensic Ballistics (A.Y. 2026-2027) across four key dimensions to develop an evidence-based Learning Enhancement Plan.

RESEARCH PROBLEM

This study aims to determine the competency-based learning challenges encountered by fourth-year Criminology students in Forensic Ballistics at PHILCST during A.Y. 2026–2027. Specifically, it seeks to answer the following questions:

1. Learning Challenges: What learning challenges do Criminology students encounter in studying Forensic Ballistics in terms of:
    - A. Course Content
    - B. Laboratory Equipment and Learning
    - C. Laboratory Activities
    - D. Instructional Strategies and Learning Support
2. Contributing Factors: What factors contribute to the learning challenges experienced by students in Forensic Ballistics?
3. Proposed Intervention: Based on the findings, what can be proposed to improve competency-based learning in Forensic Ballistics?

OBJECTIVES

To determine the competency-based learning challenges encountered by fourth-year Criminology students in studying Forensic Ballistics at PHILCST during Academic Year 2026–2027.

Specifically, this study seeks to achieve the following objectives:

1. To identify the learning challenges students encounter in Forensic Ballistics in terms of:
    - Course Content
    - Laboratory Equipment and Learning
    - Laboratory Activities
    - Instructional Strategies and Learning Support
2. To determine the factors that contribute to these learning challenges.
3. To propose an evidence-based Competency-Based Learning Enhancement Plan based on the findings.

SCOPE AND DELIMITATIONS

- **Locale:** Philippine College of Science and Technology (PHILCST), Calasiao, Pangasinan
- **Respondents:** Fourth-year BS Criminology students enrolled during A.Y. 2026–2027
- **Focus:** Competency-based learning challenges in Forensic Ballistics across four dimensions:
  - Course Content
  - Laboratory Equipment and Learning
  - Laboratory Activities
  - Instructional Strategies and Learning Support
- Output: A proposed Competency-Based Learning Enhancement Plan

- Excludes students from other year levels or institutions
- Does not evaluate the actual performance of law enforcement personnel
- Does not conduct technical examinations of firearms and ballistic evidence
- Findings are based solely on data gathered from the selected respondents at PHILCST

THEORETICAL FRAMEWORK

1. **Experiential Learning Theory.** Developed by David A. Kolb; updated by Kolb and Kolb (2022). Explains how learning is a process where knowledge is developed through experience, reflection, understanding, and application. In this study, it helps explain how students develop competencies in Forensic Ballistics through theoretical concepts and practical laboratory experiences across four stages: Concrete Experience, Reflective Observation, Abstract Conceptualization, and Active Experimentation.
2. **Competency-Based Learning.** Supported by Stewart (2021) and Wang et al. (2021). Emphasizes the development and demonstration of specific knowledge and skills rather than mere completion of lessons or examinations. It helps explain the learning outcomes, instructional strategies, feedback, and assessment practices expected in Forensic Ballistics.

Integration: These theories help explain the learning challenges experienced by Criminology students in Forensic Ballistics and the instructional support needed to develop their competencies.

CONCEPTUAL FRAMEWORK

INPUT

Learning Challenges in Forensic Ballistics

1. Course Content
2. Laboratory Equipment and Learning Resources
3. Laboratory Activities
4. Instructional Strategies and Learning Support

Contributing Factors

- Difficulties in understanding concepts
- Availability and accessibility of learning resources
- Opportunities for practical learning
- Instructional guidance and feedback

PROCESS

Research Procedures

1. Development of the research questionnaire
2. Validation of the research instrument
3. Administration of the questionnaire
4. Collection and organization of data
5. Statistical analysis and interpretation of findings
6. Identification of major learning challenges and contributing factors

OUTPUT

Proposed Competency-Based Learning Enhancement Plan in Forensic Ballistics for Criminology Students

- Recommended improvements in instructional strategies
- Enhanced laboratory learning experiences
- Improved access to learning resources
- Strengthened learning support
- Development of students' forensic ballistics competencies

IMPORTANCE OF THE STUDY

This study may benefit the following:

- **Criminology Students:** May help them recognize the challenges they encounter in learning Forensic Ballistics and identify areas where they need additional academic support.
- **Instructors:** May help them understand students' learning difficulties and improve their teaching strategies, feedback, and practical laboratory activities.
- **Philippine College of Science and Technology (PHILCST):** May provide useful information for improving laboratory facilities, learning materials, and educational programs related to Forensic Ballistics.
- **Future Researchers:** May serve as a reference for future studies on competency-based learning and forensic education.

RESEARCH METHODOLOGY

- **Research Design:** Descriptive-Quantitative Research Design
  - Aims to identify, describe, and measure the learning challenges experienced by students.
- **Locale of the Study:** Philippine College of Science and Technology (PHILCST), Calasiao, Pangasinan
- **Respondents:** Fourth-year BS Criminology students who have taken Forensic Ballistics during A.Y. 2026–2027 (Target: 209 students)
  - **Sampling:** Total enumeration or random sampling
  - One Forensic Ballistics instructor will serve as a key informant for supplementary data.
- **Research Instrument:** Researcher-made Questionnaire (Likert Scale)
  - **Part I:** Respondents' Profile
  - **Part II:** Learning Challenges (Course Content, Lab Equipment, Lab Activities, Instructional Support)
  - **Part III:** Contributing Factors and Proposed Improvements
- **Data Analysis:** Frequency, Percentage, Weighted Mean, and Ranking

PROPOSED DATA GATHERING TOOL

Instrument: Researcher-Made Questionnaire (Likert Scale)

Part I: Respondents' Profile

- Academic information directly relevant to the study (e.g., year level)
- Only necessary information collected to protect privacy

Part II: Learning Challenges in Forensic Ballistics

- Measures challenges across four dimensions:
  - Course Content
  - Laboratory Equipment and Learning
  - Laboratory Activities
  - Instructional Strategies and Learning Support

Part III: Contributing Factors and Proposed Improvements

- Closed-ended questions for quantitative analysis
- Optional open-ended questions grouped by recurring themes

Validation & Scale:

- Reviewed by research adviser and qualified validators for clarity, relevance, and alignment
- 5-Point Likert Scale: 4.21–5.00 (Very High) to 1.00–1.80 (Very Low)

Supplementary Tool: Interview Guide for the Forensic Ballistics instructor (analyzed through thematic summarization)

DATA GATHERING PROCEDURE

Step 1: Instrument Preparation

- Prepare the researcher-made questionnaire based on the objectives and research problems.
- Submit to the research adviser and qualified validators for evaluation and revision.

Step 2: Securing Permission

- Obtain authorization from PHILCST school authorities.
- Coordinate with the Forensic Ballistics instructor and authorized personnel for scheduling.

Step 3: Informed Consent

- Explain the purpose of the study, participation requirements, and respondents' rights.
- Secure informed consent; emphasize voluntary participation and confidentiality.

Step 4: Administration of Questionnaire

- Distribute questionnaires to selected respondents (or all qualified students if total enumeration is used).
- Give sufficient time to answer independently and honestly without researcher influence.

Step 5: Collection and Checking

- Collect completed questionnaires and check for completeness.
- Code, organize, and tabulate responses for statistical analysis.

Step 6: Supplementary Interview (if approved)

- Conduct a separate interview with the Forensic Ballistics instructor using an approved interview guide.
- Summarize instructor's responses separately from student survey statistics.

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ETHICAL CONSIDERATION

- Permission to Conduct the Study
- Informed Consent and Voluntary Participation
- Confidentiality and Privacy
- Secure Data Management
- Honesty and Accuracy in Reporting
- Protection from Harm or Pressure
- Proper Use of Research Sources

CONCLUSION

- This study will identify and measure the specific competency-based learning challenges encountered by fourth-year Criminology students in Forensic Ballistics across four dimensions: course content, laboratory equipment and learning, laboratory activities, and instructional strategies and learning support.
- The findings will determine the factors contributing to these challenges and rank them according to their severity using weighted mean and ranking.
- Ultimately, the study will produce an evidence-based Competency-Based Learning Enhancement Plan to improve instruction, laboratory learning experiences, and academic support in Forensic Ballistics at PHILCST.

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THANK YOU